



MAKERERE UNIVERSITY

COLLEGE OF COMPUTING AND INFORMATION SCIENCES
BSE4104: EMERGING TRENDS IN SOFTWARE ENGINEERING

AI-NATIVE & AGENTIC ENGINEERING CAPSTONE: WEEK 3 SUBMISSION

WEEK 3 RAG FAILURE ANALYSIS

PROJECT: UNIVERSITY STUDENT-SUPPORT CASE AGENT

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Week 3 Retrieval and Grounding Failures

F1: Persisted embedding provider mismatch

Observed: /chat failed before model invocation with Chroma's embedding-function conflict: the existing collection was persisted with onnx_mini_lm_l6_v2, while runtime configuration selected gemini-embedding.

Cause: Chroma stores the embedding-function identity in the collection. Opening the collection with a different function raises instead of silently switching providers.

Resolution: src/indexer.py now falls back to get_collection() for an existing collection when the conflict is specifically an embedding-function mismatch. Rebuilds with --force still use the explicitly selected provider.

Evidence: tests/test_retrieval.py passes 10/10 after the fix.

F2: Excluded source and stale evaluation mapping

Observed: the prompt specification and the test file disagree about several expected sources. The current test expects D01/D02 for some conduct questions, although both are excluded or unavailable; the v2.0 prompt map uses the active corpus documents for its R1-R8 scenarios. D01 is an image-only scan and D02 is unavailable, so those expectations cannot be satisfied by the current indexed text.

Cause: the evaluation cases were not updated when the corpus was repointed from the earlier placeholder set to the fetched corpus.

Impact: a failing citation assertion would not reliably indicate a retrieval defect until the test map and source register are made consistent.

F3: False-positive nearest-neighbour evidence (fixed)

Observed: the deliberately unanswerable question What are the sandwich-degree fees? retrieved three D11 chunks, with the best observed distance about 0.490. Similar out-of-scope questions also retrieved D11.

Cause: src/retriever.py always returns the top-k nearest chunks and has no configured relevance threshold or query classification gate.

Impact: the model can receive evidence for a question the corpus does not cover, weakening the no-fabrication and no-citation requirements.

Resolution: src/retriever.py now applies an unsupported-request guard and a lexical evidence-coverage gate before returning chunks. The 15-case local RAG evaluation passes after this change.

F4: Small-document under-ranking (fixed)

Observed: the CoCIS programmes question R6 returned D11 before D07. D07 is only 57 words, while D11 is a large policy manual with broad academic vocabulary.

Cause: the current persisted ONNX nearest-neighbour index is not sufficiently discriminative for a very short source document.

Impact: the correct source may appear in top-k but not be selected as the primary evidence.

Resolution: retrieval now searches a wider candidate pool, supplements vector results with exact lexical metadata matches, and prioritizes title/heading matches. R6 now retrieves D07 and is covered by the passing RAG evaluation.